CSE 464 Project Part 1 Readme

GitHub Repository: https://github.com/aboshra11/CSE-464-2026-aboshra

Project description: This project implements a Java-based directed graph parser and graph manipulation tool for graphs written in DOT format. The program supports parsing DOT files, adding nodes, adding edges, exporting graph information to text, exporting the graph to DOT format, and generating PNG graph images using Graphviz. The project also includes unit tests and supports Maven builds using the command mvn package.

Tools and Technologies Used:

- Java JDK 21
- IntelliJ IDEA
- Graphviz
- Apache Maven
- Junit
- GitHub

How to run the project:

1. Clone the GitHub repository.

2. Navigate to the project folder.

3. Make sure Java, Maven, and Graphviz are installed.

4. Build the project using:

   mvn package

5. Run Main.java to execute the program.

The program will:

- parse the DOT graph

- print graph information

- save graph_output.txt

- generate exported_graph.dot

- generate exported_graph.png

Implemented Features:

Feature 1

parseGraph(String filepath)

toString()

outputGraph(String filepath)

Feature 2

addNode(String label)

addNodes(String[] labels)

Feature 3

addEdge(String srcLabel, String dstLabel)

Feature 4

outputDOTGraph(String path)

outputGraphics(String path, String format)

Example Input (input.dot):

```
digraph {
a -> b;
b -> c;
c -> d;
d -> a;
a -> e;
```

e -> f;

e -> g;

f -> h;

g -> h;

}

Example output:

Number of nodes: 12

Node labels: a b c d e f g h x y z m

Number of edges: 12

Edges:

a -> b

b -> c

c -> d

d -> a

a -> e

e -> f

e -> g

f -> h

g -> h

x -> y

y -> z

m -> a

GitHub commit links:

Feature 1 commit: Parse DOT graph and create directed graph object

https://github.com/aboshra11/CSE-464-2026-aboshra/commit/30afacb3955193ff1730c508db37b6ef555292f5

Feature 2 commit: Add node and add multiple nodes APIs

https://github.com/aboshra11/CSE-464-2026-aboshra/commit/cfd5def2cd565ddbb1a905cd7e61204ba3e51599

Feature 3 commit: Add edge API with duplicate edge checking

https://github.com/aboshra11/CSE-464-2026-aboshra/commit/7328460d1956d6beb1be07f01616e75ae0b5d5b9

Feature 4 commit: Feature 4: Output graph to DOT file and PNG graphics

https://github.com/aboshra11/CSE-464-2026-aboshra/commit/d12ce03b1e37be13a7fcec7863936d5113f5e115

Tests commit: Add JUnit tests for graph parser features

https://github.com/aboshra11/CSE-464-2026-aboshra/commit/8b648596f64c2da3b63184f1efc1a4271d0298ab

Screenshots:

Feature 1 (Parse DOT graph):

Feature 2 adding nodes:

```
"C:\Program Files\Java\jdk-21.0.10\bin\java.exe" "-javaagent:D:\CSE 464\Project Part 1\IntelliJ IDEA 2025.3.3\lib\idea_rt.jar=49299" -Dfile.encoding
Number of nodes: 12
Node labels: a b c d e f g h x y z m
Number of edges: 9
Edges:
a -> b
b -> c
c -> d
d -> a
a -> e
e -> f
e -> g
f -> h
g -> h


Process finished with exit code 0
```

Feature 3 adding edges:

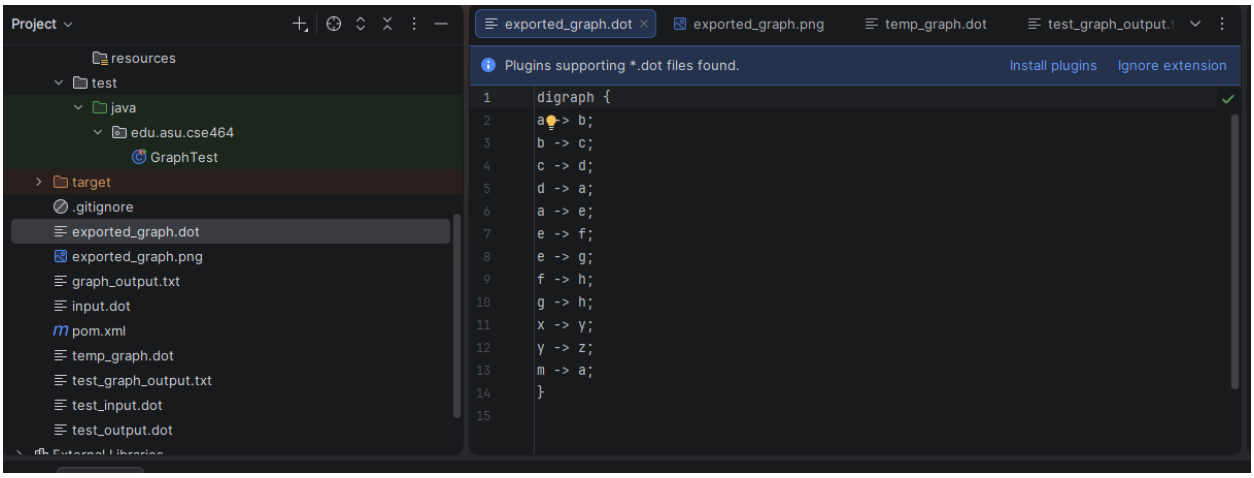
```
"C:\Program Files\Java\jdk-21.0.10\bin\java.exe" "-javaagent:D:\CSE 464\Project Part 1\IntelliJ IDEA 2025.3.3\lib\idea_rt.jar=62218" -Dfile.encoding=
Number of nodes: 12
Node labels: a b c d e f g h x y z m
Number of edges: 12
Edges:
a -> b
b -> c
c -> d
d -> a
a -> e
e -> f
e -> g
f -> h
g -> h
x -> y
y -> z
m -> a


Process finished with exit code 0
```

Feature 4 DOT + PNG export:

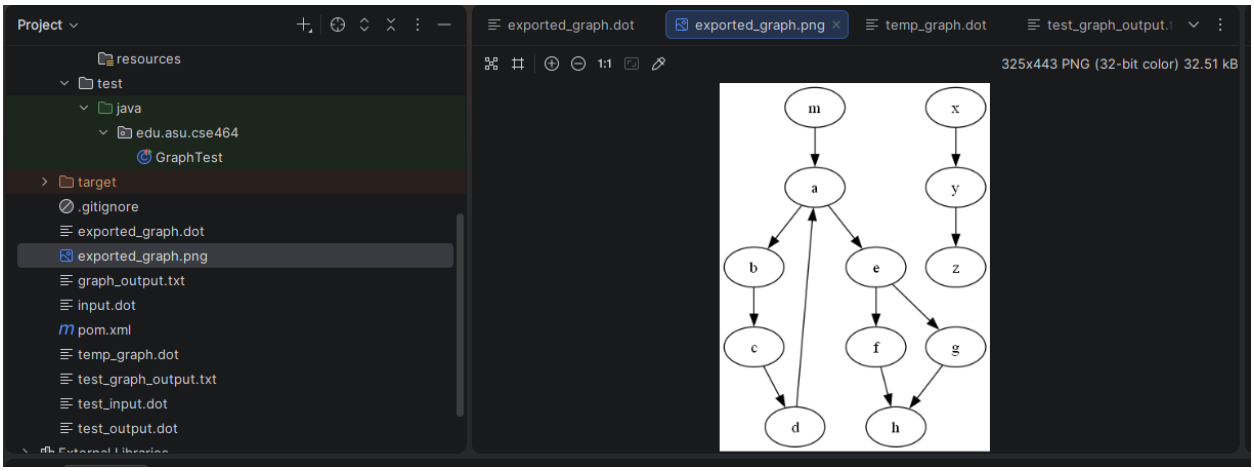
```
"C:\Program Files\Java\jdk-21.0.10\bin\java.exe" "-javaagent:D:\CSE 464\Project Part 1\IntelliJ IDEA 2025.3.3\lib\idea_rt.jar=51240" -Dfile.encoding=
Number of nodes: 12
Node labels: a b c d e f g h x y z m
Number of edges: 12
Edges:
a -> b
b -> c
c -> d
d -> a
a -> e
e -> f
e -> g
f -> h
g -> h
x -> y
y -> z
m -> a

Graph output saved to graph_output.txt
DOT graph saved to exported_graph.dot
PNG graph saved to exported_graph.png

Process finished with exit code 0
```

Graph_output.txt

Exported_graph.dot



Exported_graph.png



Tests passing:

Maven build success:

```
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Dell>  cd C:\Users\Dell\IdeaProjects\CSE464-GraphParser
PS C:\Users\Dell\IdeaProjects\CSE464-GraphParser> mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< edu.asu:CSE464-2026-aboshra >---------------------
[INFO] Building CSE464-2026-aboshra 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ CSE464-2026-aboshra ---
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ CSE464-2026-aboshra ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ CSE464-2026-aboshra ---
[INFO] skip non existing resourceDirectory C:\Users\Dell\IdeaProjects\CSE464-GraphParser\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ CSE464-2026-aboshra ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ CSE464-2026-aboshra ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running edu.asu.cse464.GraphTest
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.065 s -- in edu.asu.cse464.GraphTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ CSE464-2026-aboshra ---
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.048 s
[INFO] Finished at: 2026-03-07T17:10:17-07:00
[INFO] ------------------------------------------------------------------------
PS C:\Users\Dell\IdeaProjects\CSE464-GraphParser>
```